Math 5601: Introduction to Numerical Analysis
Homework assignment 9

Show all relevant work in detail to justify your conclusions. Partial credit depends upon the work you show. For each numerical experiment, all the .m files of your Matlab code should be electronically submitted to qingguohong@mst.edu together with a .txt file which copies all the information in the Matlab command window when you run the code to obtain the numerical results.

Problem #1: Consider the matrix $A^{(2)}$ for the second step of Gauss elimination of a non-singular matrix A in the lecture slides. Prove that there exists a s such that $a_{s2}^{(2)} \neq 0$ and $2 \leq s \leq n$. (Hint: If the fundamental row operations are applied to a non-singular matrix, then the resulted matrix is still non-singular.)

Problem #2: Prove that the SOR method

$$x_i^{(k+1)} = (1-\sigma)x_i^{(k)} + \frac{\sigma}{a_{ii}}\left\{ b_i - \sum_{j=1}^{i-1} a_{ij}x_j^{(k+1)} - \sum_{j=i+1}^{n} a_{ij}x_j^{(k)} \right\},\ i = 1, \cdots, n,\ k = 0, 1, 2, \cdots$$

is an iterative method with $M = L + \frac{1}{\sigma}D$ and $N = -\left(U + D - \frac{1}{\sigma}D\right)$. That is

$$\begin{aligned} \vec{x}^{(k+1)} &= M^{-1}N\vec{x}^{(k)} + M^{-1}\vec{b} \\ &= -\left(L + \frac{1}{\sigma}D\right)^{-1}\left(U + D - \frac{1}{\sigma}D\right)\vec{x}^{(k)} + \left(L + \frac{1}{\sigma}D\right)^{-1}\vec{b},\ k = 0, 1, 2, \cdots \end{aligned}$$

Problem #3: Consider

$$\begin{aligned} x_1 + x_3 &= 0, \\ -x_1 + x_2 &= 0, \\ x_1 + 2x_2 - 3x_3 &= 0. \end{aligned}$$

(a) Program both Jacobi and Gauss-Seidel methods. Starting from the initial guess $\vec{x}^{(0)} = (1, 1, 1)^T$, Solve the above system numerically.

(b) What behavior do you observe?

Math 5601 - Homework 09 - Tyler Glidden

Problem 1

Prove $\exists$ an s s.t. $a_{s2} \neq 0$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & & & \\ \vdots & & & \\ \vdots & & & \\ a_{n1} & & & a_{nn} \end{bmatrix} \qquad A^{(2)} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & \cdots & a_{nn} \end{bmatrix}$$

Proof: By contradiction.

Assume that there is no s s.t. $a_{s2} \neq 0$ $(2 \leq s \leq n)$. We continue the row operations until $A^{(n-1)} = \begin{pmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ 0 & 0 & a_{23} & \times & \times \\ \vdots & \vdots & a_{33} & \times & \times \\ \vdots & \vdots & 0 & \times & \times \\ 0 & 0 & 0 & 0 & \times a_{nn} \end{pmatrix}$

(A^{n-1}) is upper diagonal with $a_{22} = 0$.

Since $A^{(n-1)}$ is upper diagonal, $\det(A^{n-1}) = a_{11} \cdot \underbrace{a_{22}}_{0} \cdots a_{nn} = 0$.

$\Rightarrow$ A is singular $(!)$

Therefore, A is non-singular

Problem 2

Prove that SOR Method

$$x_i^{(k+1)} = (1-\sigma)\, x_i^{(k)} + \frac{\sigma}{a_{ii}} \left\{ b_i - \sum_{j=1}^{i-1} a_{ij} x_j^{k+1} - \sum_{j=i+1}^{n} a_{ij} x_j^{(k)} \right\} \quad i = 1, \ldots, n \quad k = 0, 1, \ldots$$

is an iterative method with $M = L + \frac{1}{\sigma} D \quad N = -\left(U + D - \frac{1}{\sigma} D\right)$

$$\vec{x}^{(k+1)} = M^{-1} N \vec{x}^{(k)} + M^{-1} \vec{b}$$

Start With:

$$x_i^{k+1} = (1-\sigma) x_i^{(k)} + \frac{\sigma}{a_{ii}} \left\{ b_i - \sum_{j=1}^{i-1} a_{ij} x_j^{(k+1)} - \sum_{j=i+1}^{n} a_{ij} x_j^{(k)} \right\}$$

$$\Leftrightarrow a_{ii} x_i^{k+1} = (1-\sigma)\, a_{ii} x_i^{k} + \sigma b_i - \sigma \sum_{j=1}^{i-1} a_{ij} x_j^{k+1} - \sigma \sum_{j=i+1}^{n} a_{ij} x_j^{(k)}$$

$$\Leftrightarrow \underbrace{a_{ii}}_{D} x_i^{k+1} + \sigma \underbrace{\sum_{j=1}^{i-1}}_{L} a_{ij} x_j^{k+1} = (1-\sigma)\, a_{ii} x_i^{k} + \sigma b_i - \sigma \underbrace{\sum_{j=i+1}}_{U} a_{ij} x_j^{(k)}$$

$$\Leftrightarrow \frac{D}{\sigma} x^{k+1} + \sigma \frac{L}{\sigma} \vec{x}^{k+1} = \frac{(1-\sigma)}{\sigma} D \vec{x}^{k} + \frac{\sigma \vec{b}}{\sigma} - \frac{U x^{(k)}}{\sigma}$$

$$\Leftrightarrow \left(\frac{1}{\sigma} D + L\right) \vec{x}^{k+1} = \left(\frac{1}{\sigma} D - D - U\right) x^{k} + \vec{b}$$

$$\Leftrightarrow \vec{x}^{k+1} = \left(\frac{1}{\sigma} D + L\right)^{-1} (-1) \left(D - \frac{1}{\sigma} D + U\right) \vec{x}^{k} + \left(\frac{1}{\sigma} D + L\right)^{-1} \vec{b}$$

This is an iterative method with

$$M = L + \frac{1}{\sigma} D, \quad N = -\left(U + D - \frac{1}{\sigma} D\right),$$

and $x^{k+1} = M^{-1} N \vec{x}^{k} + M^{-1} \vec{b}$

Where $A = M - N$

$$A = L + \frac{1}{\sigma} D + U + D - \frac{1}{\sigma} D$$

$$A = L + U + D \quad \blacksquare$$

Problem 3

a) Jacobi and GS.

b.) What behavior do you observe?

$$A\vec{x} = \vec{b}: \quad \begin{pmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \\ 1 & 2 & -3 \end{pmatrix} \vec{x} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Jacobi

100 iterations.

Since max iterations was set at 100, it did not converge.

See matlab txt file. I tried:

$10^{-6}, 10^{-5}, 10^{-4}, \ldots 10^{-1}$

and no convergence.

GS

100 iterations.